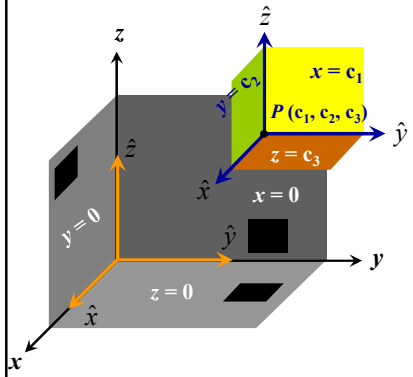


Curvilinear Coordinate Systems

Cartesian Coordinate Systems (x, y, z)



Length element:
 $d\vec{l} = dx \hat{x} + dy \hat{y} + dz \hat{z}$

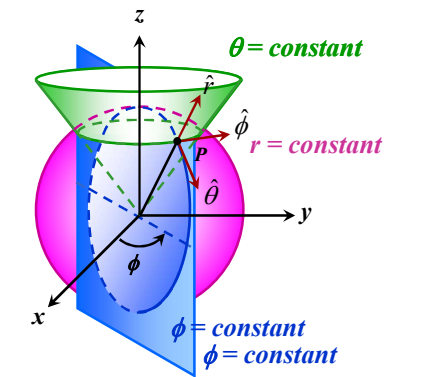
Area element: $dx dy \hat{z}$
 $dy dz \hat{x}$
 $dz dx \hat{y}$
 $d\vec{a} = dy dz \hat{x} + dz dx \hat{y} + dx dy \hat{z}$

Volume element:
 $d\tau = dx dy dz$

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Curvilinear Coordinate Systems

Spherical Polar Coordinate Systems (r, θ, ϕ)



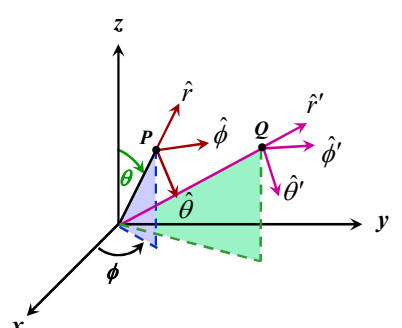
$\theta = \text{constant}$
 $r = \text{constant}$
 $\phi = \text{constant}$

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Curvilinear Coordinate Systems

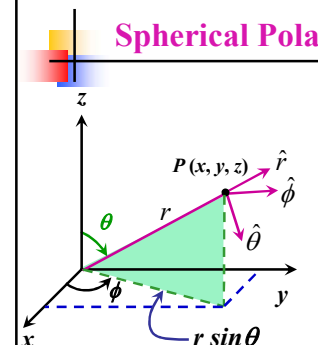
Spherical Polar Coordinate Systems (r, θ, ϕ)

$r: 0 \rightarrow \infty$
 $\theta: 0 \rightarrow \pi$
 $\phi: 0 \rightarrow 2\pi$



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Spherical Polar Coordinate Systems



$x = r \sin \theta \cos \phi = x(r, \theta, \phi)$
 $y = r \sin \theta \sin \phi = y(r, \theta, \phi)$
 $z = r \cos \theta = z(r, \theta, \phi)$
 $r = (x^2 + y^2 + z^2)^{1/2} = r(x, y, z)$
 $\sqrt{x^2 + y^2} = r \sin \theta (\cos^2 \phi + \sin^2 \phi)^{1/2}$
 $\theta = \tan^{-1} \left(\sqrt{x^2 + y^2} / z \right) = \theta(x, y, z)$
 $\phi = \tan^{-1} (y/x) = \phi(x, y, z)$
 $\vec{r} = r \sin \theta \cos \phi \hat{x} + r \sin \theta \sin \phi \hat{y} + r \cos \theta \hat{z}$

$\frac{\partial \vec{r}}{\partial r} = \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z} \rightarrow \hat{r}$
 $\frac{\partial \vec{r}}{\partial \theta} = r \cos \theta \cos \phi \hat{x} + r \cos \theta \sin \phi \hat{y} - r \sin \theta \hat{z} \rightarrow \frac{1}{r} \frac{\partial \vec{r}}{\partial \theta} = \hat{\theta}$
 $\frac{\partial \vec{r}}{\partial \phi} = -r \sin \theta \sin \phi \hat{x} + r \sin \theta \cos \phi \hat{y} \rightarrow \frac{1}{r \sin \theta} \frac{\partial \vec{r}}{\partial \phi} = \hat{\phi}$

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Spherical Polar Coordinate Systems

$$\hat{r} = \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z}$$

$$\hat{\theta} = \cos \theta \cos \phi \hat{x} + \cos \theta \sin \phi \hat{y} - \sin \theta \hat{z}$$

$$\hat{\phi} = -\sin \phi \hat{x} + \cos \phi \hat{y}$$

$$\begin{pmatrix} \hat{r} \\ \hat{\theta} \\ \hat{\phi} \end{pmatrix} = \begin{pmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix}$$

$$\begin{pmatrix} \hat{r} \\ \hat{\theta} \\ \hat{\phi} \end{pmatrix} = \mathbf{A} \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix} \longrightarrow \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix} = \mathbf{A}^{-1} \begin{pmatrix} \hat{r} \\ \hat{\theta} \\ \hat{\phi} \end{pmatrix}$$

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Spherical Polar Coordinate Systems

$$\hat{r} = \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z}$$

$$\hat{\theta} = \cos \theta \cos \phi \hat{x} + \cos \theta \sin \phi \hat{y} - \sin \theta \hat{z}$$

$$\hat{\phi} = -\sin \phi \hat{x} + \cos \phi \hat{y}$$

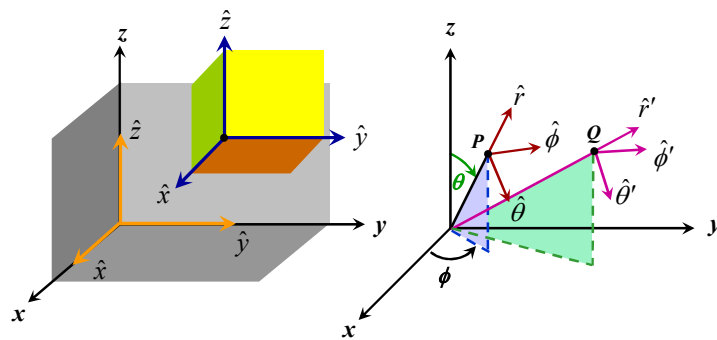
$$\frac{\partial \hat{r}}{\partial r} = 0$$

$$\frac{\partial \hat{r}}{\partial \theta} = \cos \theta \cos \phi \hat{x} + \cos \theta \sin \phi \hat{y} - \sin \theta \hat{z} = \hat{\theta} \neq 0$$

$$\frac{\partial \hat{r}}{\partial \phi} = -\sin \theta \sin \phi \hat{x} + \sin \theta \cos \phi \hat{y} = \hat{\phi} \neq 0$$

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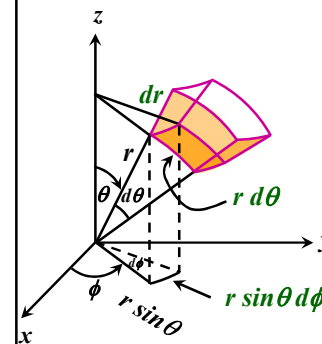
Curvilinear Coordinate Systems



Be careful while integrating/differentiating $(\hat{r}, \hat{\theta}, \hat{\phi})$

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Spherical Polar Coordinate Systems



Length element: $d\vec{l} = ?$

$$d\vec{l} = dx \hat{x} + dy \hat{y} + dz \hat{z} \quad \text{in CCS}$$

$$d\vec{l} = dl_r \hat{r} + dl_\theta \hat{\theta} + dl_\phi \hat{\phi}$$

$$= dr \hat{r} + d\theta \hat{\theta} + d\phi \hat{\phi}$$

$$d\vec{l} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$$

Area element: $r dr d\theta \hat{\phi}$

$$r \sin \theta dr d\phi \hat{\theta}$$

$$r^2 \sin \theta d\theta d\phi \hat{r}$$

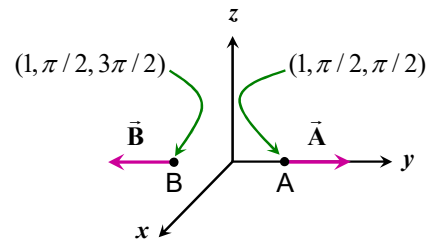
$$d\vec{a} = r^2 \sin \theta d\theta d\phi \hat{r} + r \sin \theta dr d\phi \hat{\theta} + r dr d\theta \hat{\phi}$$

Volume element:

$$d\tau = r^2 \sin \theta dr d\theta d\phi$$

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Understanding



In CCS

$$\vec{A} = \hat{y}$$

$$\vec{B} = -\hat{y}$$

$$\vec{A} + \vec{B} = \vec{0}$$

$$\vec{A} \cdot \vec{B} = -1$$

In SPCS

$$\vec{A} = \hat{r}$$

$$\vec{B} = \hat{r}$$

$$\vec{A} + \vec{B} = 2\hat{r}$$

$$\vec{A} \cdot \vec{B} = 1$$

$$\hat{r} = \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z}$$

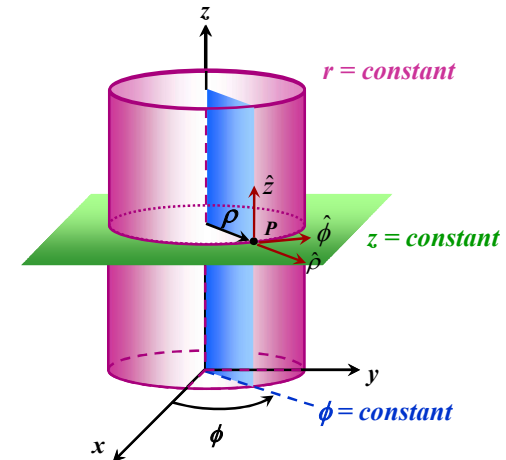
$$\hat{r}_A = \hat{y}$$

$$\hat{r}_B = -\hat{y}$$

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Curvilinear Coordinate Systems

Cylindrical Polar Coordinate Systems (ρ, ϕ, z)



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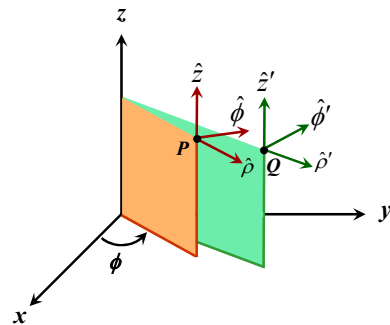
Curvilinear Coordinate Systems

Cylindrical Polar Coordinate Systems (ρ, ϕ, z)

$$\rho: 0 \rightarrow \infty$$

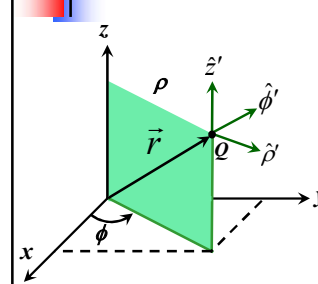
$$\phi: 0 \rightarrow 2\pi$$

$$z: -\infty \rightarrow +\infty$$



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Cylindrical Polar Coordinate Systems



$$x = \rho \cos \phi = x(\rho, \phi, z)$$

$$y = \rho \sin \phi = y(\rho, \phi, z)$$

$$z = z = z(\rho, \phi, z)$$

$$\rho = (x^2 + y^2)^{1/2} = \rho(x, y, z)$$

$$\phi = \tan^{-1}(y/x) = \phi(x, y, z)$$

$$z = z = z(x, y, z)$$

$$\vec{r} = \rho \cos \phi \hat{x} + \rho \sin \phi \hat{y} + z \hat{z}$$

$$\frac{\partial \vec{r}}{\partial \rho} = \cos \phi \hat{x} + \sin \phi \hat{y} \rightarrow \hat{\rho}$$

$$\frac{\partial \vec{r}}{\partial \phi} = -\rho \sin \phi \hat{x} + \rho \cos \phi \hat{y} \rightarrow \frac{1}{\rho} \frac{\partial \vec{r}}{\partial \phi} = \hat{\phi}$$

$$\frac{\partial \vec{r}}{\partial z} = \hat{z} \rightarrow \hat{z}$$

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Cylindrical Polar Coordinate Systems

$$\hat{\rho} = \cos \varphi \hat{x} + \sin \varphi \hat{y}$$

$$\hat{\phi} = -\sin \varphi \hat{x} + \cos \varphi \hat{y}$$

$$\hat{z} = \hat{z}$$

$$\begin{pmatrix} \hat{\rho} \\ \hat{\phi} \\ \hat{z} \end{pmatrix} = \begin{pmatrix} \cos \varphi & \sin \varphi & 0 \\ -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix}$$

$$\begin{pmatrix} \hat{\rho} \\ \hat{\phi} \\ \hat{z} \end{pmatrix} = \mathbf{A} \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix} \longrightarrow \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix} = \mathbf{A}^{-1} \begin{pmatrix} \hat{\rho} \\ \hat{\phi} \\ \hat{z} \end{pmatrix}$$

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Cylindrical Polar Coordinate Systems

$$\hat{\rho} = \cos \varphi \hat{x} + \sin \varphi \hat{y}$$

$$\hat{\phi} = -\sin \varphi \hat{x} + \cos \varphi \hat{y}$$

$$\hat{z} = \hat{z}$$

$$\frac{\partial \hat{\rho}}{\partial r} = 0$$

$$\frac{\partial \hat{\phi}}{\partial \rho} = 0$$

$$\frac{\partial \hat{\rho}}{\partial \varphi} = -\sin \varphi \hat{x} + \cos \varphi \hat{y} = \hat{\phi} \neq 0$$

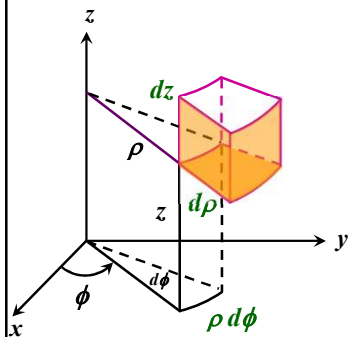
$$\frac{\partial \hat{\phi}}{\partial \varphi} = -\cos \varphi \hat{x} - \sin \varphi \hat{y} = -\hat{r} \neq 0$$

$$\frac{\partial \hat{\rho}}{\partial z} = 0$$

$$\frac{\partial \hat{\phi}}{\partial z} = 0$$

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Cylindrical Polar Coordinate Systems



Length element: $d\vec{l} = ?$

$$d\vec{l} = dx \hat{x} + dy \hat{y} + dz \hat{z} \quad \text{in CCS}$$

$$\begin{aligned} d\vec{l} &= d\vec{l}_\rho \hat{\rho} + d\vec{l}_\phi \hat{\phi} + d\vec{l}_z \hat{z} \\ &= d\rho \hat{\rho} + \rho d\phi \hat{\phi} + dz \hat{z} \end{aligned}$$

$$d\vec{l} = d\rho \hat{\rho} + \rho d\phi \hat{\phi} + dz \hat{z}$$

Area element:

$$\begin{aligned} & d\rho dz \hat{\phi} \\ & \rho d\rho d\phi \hat{z} \\ & \rho dz d\phi \hat{\rho} \end{aligned}$$

$$d\vec{a} = \rho dz d\phi \hat{\rho} + d\rho dz \hat{\phi} + \rho d\rho d\phi \hat{z}$$

Volume element:

$$d\tau = \rho d\rho d\phi dz$$

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Curvilinear Coordinate Systems: Summary

$$d\vec{l} = dx \hat{x} + dy \hat{y} + dz \hat{z} \quad \begin{matrix} x & y & z \end{matrix}$$

$$dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi} \quad \begin{matrix} r & \theta & \phi \end{matrix}$$

$$d\rho \hat{\rho} + \rho d\phi \hat{\phi} + dz \hat{z} \quad \begin{matrix} \rho & \phi & z \end{matrix}$$

$$d\vec{a} = \left. \begin{matrix} dy dz \hat{x} \\ dz dx \hat{y} \\ dx dy \hat{z} \end{matrix} \right\} \left. \begin{matrix} r^2 \sin \theta d\theta d\phi \hat{r} \\ r \sin \theta d\phi dr \hat{\theta} \\ r dr d\theta \hat{\phi} \end{matrix} \right\} \left. \begin{matrix} \rho d\phi dz \hat{\rho} \\ dz d\rho \hat{\phi} \\ \rho d\rho d\phi \hat{z} \end{matrix} \right\}$$

$$d\tau = dx dy dz$$

$$r^2 \sin \theta dr d\theta d\phi$$

$$r dr d\phi dz$$

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Curvilinear Coordinate Systems: Summary

h_1	h_2	h_3	ξ_1	ξ_2	ξ_3	h_1	h_2	h_3
$\hat{x}$	$\hat{y}$	$\hat{z}$	x	y	z	1	1	1
$\hat{r}$	$\hat{\theta}$	$\hat{\phi}$	r	θ	ϕ	1	r	$r \sin \theta$
$\hat{\rho}$	$\hat{\phi}$	$\hat{z}$	ρ	ϕ	z	1	ρ	1

$$d\vec{l} = h_1 d\xi_1 \hat{\xi}_1 + h_2 d\xi_2 \hat{\xi}_2 + h_3 d\xi_3 \hat{\xi}_3$$

$$d\vec{a} = \begin{vmatrix} dy dz \hat{x} & r^2 \sin \theta d\theta d\phi \hat{r} & \rho d\phi dz \hat{\rho} \\ dz dx \hat{y} & r \sin \theta d\phi dr \hat{\theta} & dz d\rho \hat{\phi} \\ dx dy \hat{z} & r dr d\theta \hat{\phi} & \rho d\rho d\phi \hat{z} \end{vmatrix}$$

$$d\vec{a} = h_2 h_3 d\xi_2 d\xi_3 \hat{\xi}_1 + h_3 h_1 d\xi_3 d\xi_1 \hat{\xi}_2 + h_1 h_2 d\xi_1 d\xi_2 \hat{\xi}_3$$

$$d\tau = dx dy dz$$

$$r^2 \sin \theta dr d\theta d\phi$$

$$r dr d\phi dz$$

$$d\tau = h_1 h_2 h_3 d\xi_1 d\xi_2 d\xi_3$$

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Gradient in General Curvilinear Coordinate System

Consider a scalar function $\psi(\xi_1, \xi_2, \xi_3)$

$$\vec{\nabla} \psi = \alpha_1 \hat{\xi}_1 + \alpha_2 \hat{\xi}_2 + \alpha_3 \hat{\xi}_3$$

$$d\psi = \vec{\nabla} \psi \cdot d\vec{l} = h_1 \alpha_1 d\xi_1 + h_2 \alpha_2 d\xi_2 + h_3 \alpha_3 d\xi_3$$

$$d\psi = \frac{\partial \psi}{\partial \xi_1} d\xi_1 + \frac{\partial \psi}{\partial \xi_2} d\xi_2 + \frac{\partial \psi}{\partial \xi_3} d\xi_3$$

$$\alpha_1 = \frac{1}{h_1} \frac{\partial \psi}{\partial \xi_1} \quad \alpha_2 = \frac{1}{h_2} \frac{\partial \psi}{\partial \xi_2} \quad \alpha_3 = \frac{1}{h_3} \frac{\partial \psi}{\partial \xi_3}$$

$$\vec{\nabla} \psi = \frac{1}{h_1} \frac{\partial \psi}{\partial \xi_1} \hat{\xi}_1 + \frac{1}{h_2} \frac{\partial \psi}{\partial \xi_2} \hat{\xi}_2 + \frac{1}{h_3} \frac{\partial \psi}{\partial \xi_3} \hat{\xi}_3$$

$$\vec{\nabla} \equiv \hat{\xi}_1 \frac{1}{h_1} \frac{\partial}{\partial \xi_1} + \hat{\xi}_2 \frac{1}{h_2} \frac{\partial}{\partial \xi_2} + \hat{\xi}_3 \frac{1}{h_3} \frac{\partial}{\partial \xi_3}$$

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Divergence in General Curvilinear Coordinate System

$$\vec{\nabla} \psi = \frac{1}{h_1} \frac{\partial \psi}{\partial \xi_1} \hat{\xi}_1 + \frac{1}{h_2} \frac{\partial \psi}{\partial \xi_2} \hat{\xi}_2 + \frac{1}{h_3} \frac{\partial \psi}{\partial \xi_3} \hat{\xi}_3$$

Consider a scalar function $\psi = \xi_1 \quad \psi = \xi_2 \quad \psi = \xi_3$

$$\vec{\nabla} \xi_1 = \frac{\hat{\xi}_1}{h_1} \quad \vec{\nabla} \xi_2 = \frac{\hat{\xi}_2}{h_2} \quad \vec{\nabla} \xi_3 = \frac{\hat{\xi}_3}{h_3}$$

$$\vec{\nabla} \xi_1 \times \vec{\nabla} \xi_2 = \frac{\hat{\xi}_1 \times \hat{\xi}_2}{h_1 h_2} = \frac{\hat{\xi}_3}{h_1 h_2}$$

$$\hat{\xi}_3 = h_1 h_2 \vec{\nabla} \xi_1 \times \vec{\nabla} \xi_2$$

$$\hat{\xi}_1 = h_2 h_3 \vec{\nabla} \xi_2 \times \vec{\nabla} \xi_3$$

$$\hat{\xi}_2 = h_3 h_1 \vec{\nabla} \xi_3 \times \vec{\nabla} \xi_1$$

Divergence in General Curvilinear Coordinate System

$$\hat{\xi}_1 = h_2 h_3 \vec{\nabla} \xi_2 \times \vec{\nabla} \xi_3 \quad \hat{\xi}_2 = h_3 h_1 \vec{\nabla} \xi_3 \times \vec{\nabla} \xi_1 \quad \hat{\xi}_3 = h_1 h_2 \vec{\nabla} \xi_1 \times \vec{\nabla} \xi_2$$

$$\vec{V} = V_1 \hat{\xi}_1 + V_2 \hat{\xi}_2 + V_3 \hat{\xi}_3$$

$$\vec{\nabla} \cdot \vec{V} = \vec{\nabla} \cdot (V_1 \hat{\xi}_1 + V_2 \hat{\xi}_2 + V_3 \hat{\xi}_3)$$

$$\vec{\nabla} \cdot (V_1 \hat{\xi}_1) = \vec{\nabla} \cdot (V_1 h_2 h_3 \vec{\nabla} \xi_2 \times \vec{\nabla} \xi_3)$$

$$= (V_1 h_2 h_3) \vec{\nabla} \cdot (\vec{\nabla} \xi_2 \times \vec{\nabla} \xi_3) + \vec{\nabla} (V_1 h_2 h_3) \cdot (\vec{\nabla} \xi_2 \times \vec{\nabla} \xi_3)$$

$$= (V_1 h_2 h_3) [\vec{\nabla} \xi_2 \cdot (\vec{\nabla} \times \vec{\nabla} \xi_3) - \vec{\nabla} \xi_3 \cdot (\vec{\nabla} \times \vec{\nabla} \xi_2)] + \vec{\nabla} (V_1 h_2 h_3) \cdot \frac{\hat{\xi}_1}{h_2 h_3}$$

$$= \vec{\nabla} (V_1 h_2 h_3) \cdot \frac{\hat{\xi}_1}{h_2 h_3}$$

$$= \left[\frac{1}{h_1} \frac{\partial (V_1 h_2 h_3)}{\partial \xi_1} \hat{\xi}_1 + () \hat{\xi}_2 + () \hat{\xi}_3 \right] \cdot \frac{\hat{\xi}_1}{h_2 h_3} = \frac{1}{h_1 h_2 h_3} \frac{\partial (V_1 h_2 h_3)}{\partial \xi_1}$$

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Divergence in General Curvilinear Coordinate System

$$\vec{\nabla} \cdot \vec{V} = \vec{\nabla} \cdot (V_1 \hat{\xi}_1 + V_2 \hat{\xi}_2 + V_3 \hat{\xi}_3)$$

$$\vec{\nabla} \cdot (V_1 \hat{\xi}_1) = \frac{1}{h_1 h_2 h_3} \frac{\partial (V_1 h_2 h_3)}{\partial \xi_1}$$

$$\vec{\nabla} \cdot (V_2 \hat{\xi}_2) = \frac{1}{h_1 h_2 h_3} \frac{\partial (V_2 h_3 h_1)}{\partial \xi_2}$$

$$\vec{\nabla} \cdot (V_3 \hat{\xi}_3) = \frac{1}{h_1 h_2 h_3} \frac{\partial (V_3 h_1 h_2)}{\partial \xi_3}$$

$$\vec{\nabla} \cdot \vec{V} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial (V_1 h_2 h_3)}{\partial \xi_1} + \frac{\partial (V_2 h_3 h_1)}{\partial \xi_2} + \frac{\partial (V_3 h_1 h_2)}{\partial \xi_3} \right]$$

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Curl in General Curvilinear Coordinate System

$$\vec{\nabla} \times \vec{V} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{\xi}_1 & h_2 \hat{\xi}_2 & h_3 \hat{\xi}_3 \\ \partial/\partial \xi_1 & \partial/\partial \xi_2 & \partial/\partial \xi_3 \\ h_1 V_1 & h_2 V_2 & h_3 V_3 \end{vmatrix}$$

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Laplacian in General Curvilinear Coordinate System

$$\nabla^2 \psi = \vec{\nabla} \cdot \vec{\nabla} \psi = \vec{\nabla} \cdot \vec{V} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial (V_1 h_2 h_3)}{\partial \xi_1} + \frac{\partial (V_2 h_3 h_1)}{\partial \xi_2} + \frac{\partial (V_3 h_1 h_2)}{\partial \xi_3} \right]$$

$$= \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial \xi_1} \left(\frac{1}{h_1} \frac{\partial \psi}{\partial \xi_1} h_2 h_3 \right) + \frac{\partial}{\partial \xi_2} \left(\frac{1}{h_2} \frac{\partial \psi}{\partial \xi_2} h_3 h_1 \right) + \frac{\partial}{\partial \xi_3} \left(\frac{1}{h_3} \frac{\partial \psi}{\partial \xi_3} h_1 h_2 \right) \right]$$

$$\nabla^2 \psi = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial \xi_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial \psi}{\partial \xi_1} \right) + \frac{\partial}{\partial \xi_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial \psi}{\partial \xi_2} \right) + \frac{\partial}{\partial \xi_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial \psi}{\partial \xi_3} \right) \right]$$

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Summary

$$\vec{\nabla} \psi = \frac{1}{h_1} \frac{\partial \psi}{\partial \xi_1} \hat{\xi}_1 + \frac{1}{h_2} \frac{\partial \psi}{\partial \xi_2} \hat{\xi}_2 + \frac{1}{h_3} \frac{\partial \psi}{\partial \xi_3} \hat{\xi}_3$$

$$\vec{\nabla} \cdot \vec{V} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial (V_1 h_2 h_3)}{\partial \xi_1} + \frac{\partial (V_2 h_3 h_1)}{\partial \xi_2} + \frac{\partial (V_3 h_1 h_2)}{\partial \xi_3} \right]$$

$$\vec{\nabla} \times \vec{V} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{\xi}_1 & h_2 \hat{\xi}_2 & h_3 \hat{\xi}_3 \\ \partial/\partial \xi_1 & \partial/\partial \xi_2 & \partial/\partial \xi_3 \\ h_1 V_1 & h_2 V_2 & h_3 V_3 \end{vmatrix}$$

CCS

SPCS

CPCS

h_1	h_2	h_3
1	1	1
1	r	$r \sin \theta$
1	ρ	1

$$\nabla^2 \psi = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial \xi_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial \psi}{\partial \xi_1} \right) + \frac{\partial}{\partial \xi_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial \psi}{\partial \xi_2} \right) + \frac{\partial}{\partial \xi_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial \psi}{\partial \xi_3} \right) \right]$$

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Example

Determine the flux of a vector field $\vec{F}(r) = 10 e^{-2z} (r \hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \leq z \leq 1$.

Confirm the result using the divergence theorem.

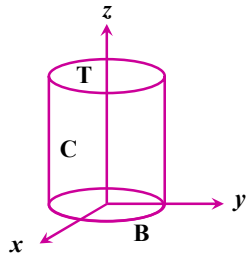
$$\text{Flux} = \oint \vec{F} \cdot d\vec{a} = \int_T \vec{F} \cdot d\vec{a} + \int_C \vec{F} \cdot d\vec{a} + \int_B \vec{F} \cdot d\vec{a}$$

For **T**, $r: 0 \rightarrow 1, \phi: 0 \rightarrow 2\pi, z = 1$

$$d\vec{a} = h_1 h_2 d\xi_1 d\xi_2 \hat{\xi}_3 = r dr d\phi \hat{z}$$

$$\vec{F} \cdot d\vec{a} = 10 e^{-2z} dr d\phi = 10 e^{-2} dr d\phi$$

$$\begin{aligned} \text{Flux from T} &= \int_{r=0}^1 \int_{\phi=0}^{2\pi} 10 e^{-2} r dr d\phi \\ &= 10 e^{-2} \int_{r=0}^1 r dr \int_{\phi=0}^{2\pi} d\phi = 10 \pi e^{-2} \end{aligned}$$



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Example

Determine the flux of a vector field $\vec{F}(r) = 10 e^{-2z} (r \hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \leq z \leq 1$.

Confirm the result using the divergence theorem.

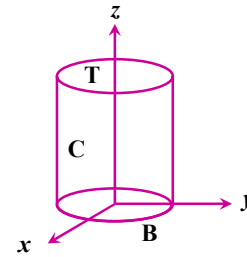
$$\text{Flux} = \oint \vec{F} \cdot d\vec{a} = \int_T \vec{F} \cdot d\vec{a} + \int_C \vec{F} \cdot d\vec{a} + \int_B \vec{F} \cdot d\vec{a}$$

For **B**, $r: 0 \rightarrow 1, \phi: 0 \rightarrow 2\pi, z = 0$

$$d\vec{a} = -r dr d\phi \hat{z}$$

$$\vec{F} \cdot d\vec{a} = -10 e^{-2z} dr d\phi = -10 dr d\phi$$

$$\begin{aligned} \text{Flux from B} &= - \int_{r=0}^1 \int_{\phi=0}^{2\pi} 10 r dr d\phi \\ &= -10 \int_{r=0}^1 r dr \int_{\phi=0}^{2\pi} d\phi = -10 \pi \end{aligned}$$



63

Example

Determine the flux of a vector field $\vec{F}(r) = 10 e^{-2z} (r \hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \leq z \leq 1$.

Confirm the result using the divergence theorem.

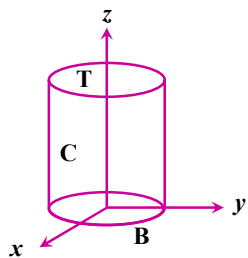
$$\text{Flux} = \oint \vec{F} \cdot d\vec{a} = \int_T \vec{F} \cdot d\vec{a} + \int_C \vec{F} \cdot d\vec{a} + \int_B \vec{F} \cdot d\vec{a}$$

For **C**, $r = 1, \phi: 0 \rightarrow 2\pi, z: 0 \rightarrow 1$

$$d\vec{a} = h_2 h_3 d\xi_2 d\xi_3 \hat{\xi}_1 = r d\phi dz \hat{r}$$

$$\vec{F} \cdot d\vec{a} = 10 e^{-2z} r^2 d\phi dz = 10 e^{-2z} d\phi dz$$

$$\begin{aligned} \text{Flux from C} &= \int_{\phi=0}^{2\pi} \int_{z=0}^1 10 e^{-2z} d\phi dz \\ &= 10 \pi (1 - e^{-2}) \end{aligned}$$



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Example

Determine the flux of a vector field $\vec{F}(r) = 10 e^{-2z} (r \hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \leq z \leq 1$.

Confirm the result using the divergence theorem.

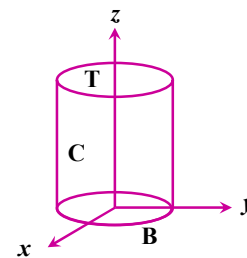
$$\text{Flux} = \oint \vec{F} \cdot d\vec{a} = \int_T \vec{F} \cdot d\vec{a} + \int_C \vec{F} \cdot d\vec{a} + \int_B \vec{F} \cdot d\vec{a}$$

$$\text{Flux from T} = 10 \pi e^{-2}$$

$$\text{Flux from B} = -10 \pi$$

$$\text{Flux from C} = 10 \pi (1 - e^{-2})$$

$$\text{Total Flux} = 0$$



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Example

Determine the flux of a vector field $\vec{F}(r) = 10 e^{-2z} (r \hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \leq z \leq 1$.

Confirm the result using the divergence theorem.

$$\begin{aligned} \vec{\nabla} \cdot \vec{F} &= \frac{1}{h_1 h_2 h_3} \left[\frac{\partial(V_1 h_2 h_3)}{\partial \xi_1} + \frac{\partial(V_2 h_3 h_1)}{\partial \xi_2} + \frac{\partial(V_3 h_1 h_2)}{\partial \xi_3} \right] \\ &= \frac{1}{r} \left[\frac{\partial(10 e^{-2z} r r)}{\partial r} + \frac{\partial(0)}{\partial \phi} + \frac{\partial(10 e^{-2z} r)}{\partial z} \right] \\ &= \frac{1}{r} \left[10 e^{-2z} \frac{\partial r^2}{\partial r} + 0 + 10 r \frac{\partial e^{-2z}}{\partial z} \right] \\ &= \frac{1}{r} [10 e^{-2z} 2r + 0 + 10 r (-2e^{-2z})] = 0 \quad \blacktriangleleft \end{aligned}$$

